

1. B

2. C

3. A

4. B

5. D

6. B

7. B

8. A

9. A

10. B

11. C

12. B.

Q. 2(i)
A. 2(i)

moles :
mass : 22.2 g
M. mass : CaCl_2

$$40 + 2(35.5)$$

$$40 + 71$$

formula : 111 g/mol

$$\text{moles} = \frac{\text{G. m}}{\text{M. m}}$$

$$= \frac{22.2}{111}$$

$$= 0.2 \text{ g/mol}$$

Q. 2(ii)
A. 2(ii)

molecules:-

$$N_A : 6.022 \times 10^{23}$$

moles:- 0.4

$$= \cancel{6.022 \times 0.4 \times 10^{23}}$$

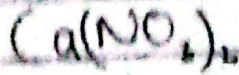
molecules:- moles $\times N_A$

$$0.4 \times 6.022 \times 10^{23}$$

$$= 2.4088 \times 10^{23}$$

Q.2(iii)

A:2(iii)



$$\begin{aligned} & 40 + 2(14) + 3(16) \\ & 40 + 28 + 48 \\ & 40 + 76 \\ & 116 \text{ g/mol} \end{aligned}$$

Q.2(iv)

A:2(iv)

molar mass :-

$$\begin{aligned} & 14 + 4(1) + (14) + 3(16) \\ & 14 + 4 + 14 + 48 \\ & 80 \text{ mol/g} \end{aligned}$$

g.mass :-

$$\begin{aligned} & \text{N} \times 2 \\ & 14 \times 2 \\ & 28 \text{ g} \end{aligned}$$

Percentage :-

$$\begin{aligned} & \frac{28}{80} \times 100\% \\ & 35\% \end{aligned}$$

35%

Q: 2(v)
A: 2(v)
(a) S in SO_3

$$x + 3(-2) = 0$$

$$x - 6 = 0$$

$$x = +6$$

(b) Cr in CrO_3

$$x + 3(-2) = 0$$

$$x - 6 = 0$$

$$x = +6$$

(c) P in P_2O_5

$$2(x) + 5(-2) = 0$$

$$2x - 10 = 0$$

$$2x = +10$$

$$x = +5$$

$$x = +5$$

Q: 2(vi)

A: 2(vi)

● mole ratio $Al : Cl_2 : AlCl_3$
 $2 : 3 : 2$

$$(a) Cl_2 = \frac{3}{2} \times 0.4 = 0.6 \text{ mol}$$

$$(b) AlCl_3 = \frac{2}{2} \times 0.4 = 0.4 \text{ mol}$$

Q: 2(vii)(06)

A: 2(vii)(08)

$$\frac{(10 \times 20) + (11 \times 80)}{100}$$

$$\frac{200 + 880}{100}$$

$$\frac{1080}{100}$$

$$10.8 \text{ amu}$$

Q: 2(viii)

A: 2(viii)

It states that it is impossible to determine, at the same time, both the position (the location) and future path (momentum) for the electron.

Therefore a definite orbit for the electron that cannot be drawn, we can only map the region where electron is likely to be found (in an orbital).

Q: 2(ii)

A: 2(ii)

Carbon dating:

Estimates the age of ~~choice~~ once-living material by comparing the C-14 activity of the sample with that of living tissue. When ${}^{14}_6\text{C} \rightarrow {}^{14}_7\text{N}$ the proton number increases from 6 to 7 (a neutron becomes a Proton) and a beta particle (electron) is emitted.

Q: 2(x/or)

A: 2(x/or)

Down Group 1:

(a) Reactivity increases because outer electron further from the nucleus, lost more easily.

(b) Density increases but melting point decreases.

Q.2 (ni) or

A:2 (ni) or

In solid NaCl the ions are locked in fixed positions in the lattice and cannot move, it does not conduct. When molten or dissolved, NaCl ionizes and the ions move freely so it conducts. Aqueous acids conduct because they ionize in water to give freely moving.

LONG

QUESTION

Q.3 (i)

A:3 (i)

(a) Relative atomic mass:

The average mass of the isotopes of an element compared with $\frac{1}{12}$ mass of a carbon-12 atom

$$Ar(U) = \frac{99.27 \times 238 + 0.72 \times 235 + 0.000 \times 234}{100}$$

$$= \frac{23626.26 + 169.2 + 0.000}{100}$$

$$\frac{23796.864}{100}$$

$$> 237.97 \text{ amu}$$



emitted is an alpha particle (helium nucleus).

$$Q: 3(ii)(c)$$

$$A: 3(ii)(c)$$

$$a) C: \frac{82.7}{12} = 6.89$$

$$H: \frac{17.3}{1} = 17.3$$

divide by 6.89

$$C = 1$$

$$H = 2.51 \times 2 = C_2H_5 \text{ (Empirical formula)}$$

$$b) \text{ Empirical mass} = 29 + 5 = 34$$

$$n = \frac{68}{34} = 2 \text{ (molecular formula)} \\ \text{(butane)}$$

Q:3(iii)
A:3(iii)

formula ΔH :

Sum of bond energies broken

Sum of bond energies formed

Bonds broken (reactants) :

$$1(\text{N} \equiv \text{N}) + 3(\text{H}-\text{H}) \\ 945 + 3(436) \Rightarrow 945 + 1308 \\ 2253 \text{ kJ}$$

Bonds formed (products) :

$$2\text{NH}_3 : 6(\text{N}-\text{H}) = 6(391) = 2346 \text{ kJ}$$

$$\Delta H = 2253 - 2346 \\ = -93 \text{ kJ/mol}$$

ΔH is negative, so the reaction is exothermic.

Molar mass :



$$\begin{aligned} & 3(12) + 8(1) \\ & 36 + 8 \\ & 44 \text{ g/mol} \end{aligned}$$

① moles $C_3H_8 = \frac{4.4}{44}$

$$= 0.1 \text{ mol}$$

② $O_2 : C_3H_8 = 5 : 1$

$$\begin{aligned} O_2 &= 5 \times 0.1 \\ &= 0.5 \text{ mol} \end{aligned}$$

③ $CO_2 : C_3H_8 = 3 : 1$

$$CO_2 : 3 \times 0.1 = 0.3 \text{ mol}$$

$$M(CO_2) = 44$$

$$\text{mass} = 0.3 \times 44 = 13.2 \text{ g}$$

④ $H_2O : C_3H_8 = 4 : 1$

$$H_2O = 0.4 \text{ mol}$$

$$\begin{aligned} \text{molecules} &= 0.4 \times 6.022 \times 10^{23} \\ &= 2.409 \times 10^{23} \end{aligned}$$